```java
class HelloWorld {
    public static void main(String[] args) {
        System.out.println("Hello, World!");
    }
}
```

```java
import java.util.Scanner;

class Calculator {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first number: ");
        double a = sc.nextDouble();

        System.out.print("Enter second number: ");
        double b = sc.nextDouble();

        System.out.println("1.Add 2.Subtract 3.Multiply 4.Divide");
        int choice = sc.nextInt();

        switch(choice) {
            case 1: System.out.println("Result = " + (a + b)); break;
            case 2: System.out.println("Result = " + (a - b)); break;
            case 3: System.out.println("Result = " + (a * b)); break;
            case 4: System.out.println("Result = " + (a / b)); break;
            default: System.out.println("Invalid Choice");
        }
    }
}
```

```java
import java.util.Scanner;

class EvenOdd {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int num = sc.nextInt();

        if(num % 2 == 0)
            System.out.println("Even");
        else
            System.out.println("Odd");
    }
}
```

```java
import java.util.Scanner;

class LeapYear {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int year = sc.nextInt();

        if((year % 4 == 0 && year % 100 != 0) || year % 400 == 0)
            System.out.println("Leap Year");
        else
            System.out.println("Not a Leap Year");
    }
}
```

```java
import java.util.Scanner;

class Table {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        for(int i = 1; i <= 10; i++) {
            System.out.println(n + " x " + i + " = " + (n * i));
        }
    }
}
```

```java
class DataTypes {
    public static void main(String[] args) {
        int a = 10;
        float b = 12.5f;
        double c = 25.78;
        char d = 'A';
        boolean e = true;

        System.out.println(a);
        System.out.println(b);
        System.out.println(c);
        System.out.println(d);
        System.out.println(e);
    }
}
```

```java
class TypeCasting {
    public static void main(String[] args) {
```

```java
        double d = 12.75;
        int i = (int)d;

        int num = 50;
        double d2 = (double)num;

        System.out.println(i);
        System.out.println(d2);
    }
}
```

```java
class Precedence {
    public static void main(String[] args) {
        int result = 10 + 5 * 2;

        System.out.println(result);
    }
}
```

```java
import java.util.Scanner;

class Grade {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int marks = sc.nextInt();

        if(marks >= 90)
            System.out.println("Grade A");
        else if(marks >= 80)
            System.out.println("Grade B");
        else if(marks >= 70)
            System.out.println("Grade C");
        else if(marks >= 60)
            System.out.println("Grade D");
        else
            System.out.println("Grade F");
    }
}
```

```java
import java.util.*;

class GuessGame {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Random rand = new Random();
```

```
        int number = rand.nextInt(100) + 1;
        int guess;

        do {
            System.out.print("Guess: ");
            guess = sc.nextInt();

            if(guess > number)
                System.out.println("Too High");
            else if(guess < number)
                System.out.println("Too Low");

        } while(guess != number);

        System.out.println("Correct!");
    }
}
```

```
import java.util.Scanner;

class Factorial {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        long fact = 1;

        for(int i = 1; i <= n; i++)
            fact *= i;

        System.out.println("Factorial = " + fact);
    }
}
```

```
class Overload {

    static int add(int a, int b) {
        return a + b;
    }

    static double add(double a, double b) {
        return a + b;
    }

    static int add(int a, int b, int c) {
        return a + b + c;
    }
```

```
    public static void main(String[] args) {
        System.out.println(add(10,20));
        System.out.println(add(10.5,20.5));
        System.out.println(add(10,20,30));
    }
}
```

```
class Fibonacci {

    static int fib(int n) {
        if(n <= 1)
            return n;

        return fib(n - 1) + fib(n - 2);
    }

    public static void main(String[] args) {
        System.out.println(fib(6));
    }
}
```

```
import java.util.Scanner;

class ArrayAverage {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        int arr[] = new int[n];
        int sum = 0;

        for(int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
            sum += arr[i];
        }

        double avg = (double)sum / n;

        System.out.println("Sum = " + sum);
        System.out.println("Average = " + avg);
    }
}
```

```
import java.util.Scanner;
```

```java
class ReverseString {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        String str = sc.nextLine();

        String rev = new StringBuilder(str).reverse().toString();

        System.out.println(rev);
    }
}
```

```java
import java.util.Scanner;

class Palindrome {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        String str = sc.nextLine()
                        .replaceAll("[^a-zA-Z0-9]", "")
                        .toLowerCase();

        String rev = new StringBuilder(str).reverse().toString();

        if(str.equals(rev))
            System.out.println("Palindrome");
        else
            System.out.println("Not Palindrome");
    }
}
```

```java
class Car {
    String make;
    String model;
    int year;

    void displayDetails() {
        System.out.println("Make: " + make);
        System.out.println("Model: " + model);
        System.out.println("Year: " + year);
    }
}

public class Main {
    public static void main(String[] args) {
```

```
        Car c1 = new Car();

        c1.make = "Toyota";
        c1.model = "Corolla";
        c1.year = 2024;

        c1.displayDetails();
    }
}
```

```
class Animal {
    void makeSound() {
        System.out.println("Animal makes sound");
    }
}

class Dog extends Animal {
    @Override
    void makeSound() {
        System.out.println("Bark");
    }
}

public class Main {
    public static void main(String[] args) {
        Animal a = new Animal();
        Dog d = new Dog();

        a.makeSound();
        d.makeSound();
    }
}
```

```
interface Playable {
    void play();
}

class Guitar implements Playable {
    public void play() {
        System.out.println("Playing Guitar");
    }
}

class Piano implements Playable {
    public void play() {
        System.out.println("Playing Piano");
    }
}
```

```java
public class Main {
    public static void main(String[] args) {
        Playable g = new Guitar();
        Playable p = new Piano();

        g.play();
        p.play();
    }
}
```

```java
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        try {
            int a = sc.nextInt();
            int b = sc.nextInt();

            int result = a / b;

            System.out.println("Result = " + result);
        }
        catch(ArithmeticException e) {
            System.out.println("Division by zero is not allowed");
        }
    }
}
```

```java
class InvalidAgeException extends Exception {
    InvalidAgeException(String msg) {
        super(msg);
    }
}

public class Main {

    static void checkAge(int age) throws InvalidAgeException {
        if(age < 18)
            throw new InvalidAgeException("Age must be 18 or above");
    }

    public static void main(String[] args) {
        try {
            checkAge(16);
```

```
        }
        catch(InvalidAgeException e) {
            System.out.println(e.getMessage());
        }
    }
}
```

```
import java.io.FileWriter;
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {

        try {
            Scanner sc = new Scanner(System.in);

            String text = sc.nextLine();

            FileWriter fw = new FileWriter("output.txt");

            fw.write(text);

            fw.close();

            System.out.println("Data written successfully");
        }
        catch(Exception e) {
            System.out.println(e);
        }
    }
}
```

```
import java.io.*;

public class Main {
    public static void main(String[] args) {

        try {
            BufferedReader br =
                new BufferedReader(new FileReader("output.txt"));

            String line;

            while((line = br.readLine()) != null) {
                System.out.println(line);
            }

            br.close();
```

```
        }
        catch(Exception e) {
            System.out.println(e);
        }
    }
}
```

```
import java.util.*;

public class Main {
    public static void main(String[] args) {

        ArrayList<String> students = new ArrayList<>();

        students.add("Rahul");
        students.add("Amit");
        students.add("Priya");

        for(String name : students)
            System.out.println(name);
    }
}
```

```
import java.util.*;

public class Main {
    public static void main(String[] args) {

        HashMap<Integer,String> map = new HashMap<>();

        map.put(101, "Rahul");
        map.put(102, "Priya");

        System.out.println(map.get(101));
    }
}
```

```
class MyThread extends Thread {

    public void run() {
        for(int i=1;i<=5;i++) {
            System.out.println("Thread Running");
        }
    }
}

public class Main {
```

```
    public static void main(String[] args) {

        MyThread t1 = new MyThread();
        MyThread t2 = new MyThread();

        t1.start();
        t2.start();
    }
}
```

```
import java.util.*;

public class Main {
    public static void main(String[] args) {

        List<String> list =
                Arrays.asList("Java","Python","C++","Go");

        Collections.sort(list,
                (a,b) -> a.compareTo(b));

        System.out.println(list);
    }
}
```

```
import java.util.*;
import java.util.stream.*;

public class Main {
    public static void main(String[] args) {

        List<Integer> nums =
            Arrays.asList(1,2,3,4,5,6,7,8);

        List<Integer> even =
            nums.stream()
                .filter(n -> n % 2 == 0)
                .collect(Collectors.toList());

        System.out.println(even);
    }
}
```

```
import java.util.List;

record Person(String name, int age) {}
```

```
public class Main {
    public static void main(String[] args) {

        List<Person> persons = List.of(
                new Person("Rahul",20),
                new Person("Priya",17)
        );

        persons.stream()
               .filter(p -> p.age() >= 18)
               .forEach(System.out::println);
    }
}
```

```
public class Main {

    static void check(Object obj) {

        switch(obj) {

            case Integer i ->
                System.out.println("Integer: " + i);

            case String s ->
                System.out.println("String: " + s);

            case Double d ->
                System.out.println("Double: " + d);

            default ->
                System.out.println("Unknown");
        }
    }

    public static void main(String[] args) {
        check(10);
        check("Java");
    }
}
```

```
import java.sql.*;

public class Main {
    public static void main(String[] args) {

        try {
            Connection con =
                DriverManager.getConnection(
```

``   "jdbc:mysql://localhost:3306/studentdb",   "root",   "password");   Statement st = con.createStatement();   ResultSet rs =   st.executeQuery("SELECT * FROM students");   while(rs.next()) {   System.out.println(rs.getInt("id")   + " "   + rs.getString("name"));   }   con.close();   }   catch(Exception e) {   System.out.println(e);   }   } } ```

```
        "jdbc:mysql://localhost:3306/studentdb",
        "root",
        "password");

      Statement st = con.createStatement();

      ResultSet rs =
        st.executeQuery("SELECT * FROM students");

      while(rs.next()) {
        System.out.println(rs.getInt("id")
                + " "
                + rs.getString("name"));
      }

      con.close();
    }
    catch(Exception e) {
      System.out.println(e);
    }
  }
}
```

```
import java.sql.*;

public class StudentDAO {

  public void insertStudent(int id,String name)
      throws Exception {

    Connection con =
      DriverManager.getConnection(
      "jdbc:mysql://localhost:3306/studentdb",
      "root","password");

    PreparedStatement ps =
      con.prepareStatement(
      "INSERT INTO students VALUES(?,?)");

    ps.setInt(1,id);
    ps.setString(2,name);

    ps.executeUpdate();
    con.close();
  }
}
```

```
import java.sql.*;
```

```
public class Main {

    public static void main(String[] args) {

        try {

            Connection con =
                DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/bank",
                "root",
                "password");

            con.setAutoCommit(false);

            Statement st = con.createStatement();

            st.executeUpdate(
                "UPDATE accounts SET balance=balance-1000 WHERE id=1");

            st.executeUpdate(
                "UPDATE accounts SET balance=balance+1000 WHERE id=2");

            con.commit();

            System.out.println("Transfer Successful");
        }
        catch(Exception e) {
            System.out.println(e);
        }
    }
}
```

```
module com.utils {
    exports com.utils;
}

module com.greetings {
    requires com.utils;
}
```

```
import java.net.*;

public class Server {
    public static void main(String[] args)
            throws Exception {

        ServerSocket ss =
```

```
            new ServerSocket(5000);

        Socket s = ss.accept();

        System.out.println("Client Connected");
    }
}

import java.net.*;

public class Client {
    public static void main(String[] args)
            throws Exception {

        Socket s =
            new Socket("localhost",5000);

        System.out.println("Connected");
    }
}
```

```
import java.net.http.*;
import java.net.URI;

public class Main {

    public static void main(String[] args)
            throws Exception {

        HttpClient client =
            HttpClient.newHttpClient();

        HttpRequest request =
            HttpRequest.newBuilder()
            .uri(URI.create(
             "https://api.github.com"))
            .build();

        HttpResponse<String> response =
            client.send(
                request,
                HttpResponse.BodyHandlers.ofString());

        System.out.println(response.statusCode());
        System.out.println(response.body());
    }
}
```

```
javac Demo.java
javap -c Demo

0: getstatic
3: ldc
5: invokevirtual
8: return
```

1. Compile Java Program.
2. Open JD-GUI or CFR.
3. Load Demo.class.
4. View Decompiled Source.

```java
import java.lang.reflect.*;

class Demo {
    public void display() {
        System.out.println("Hello");
    }
}

public class Main {

    public static void main(String[] args)
            throws Exception {

        Class<?> c = Class.forName("Demo");

        Method m =
            c.getDeclaredMethod("display");

        Object obj =
            c.getDeclaredConstructor()
             .newInstance();

        m.invoke(obj);
    }
}
```

```java
public class Main {

    public static void main(String[] args) {

        for(int i=1;i<=100000;i++) {

            Thread.startVirtualThread(() ->
                System.out.println(
                "Virtual Thread"));
```

```java
        }
    }
}
```

```java
import java.util.concurrent.*;

public class Main {

    public static void main(String[] args)
            throws Exception {

        ExecutorService service =
            Executors.newFixedThreadPool(3);

        Callable<Integer> task =
            () -> 10 + 20;

        Future<Integer> future =
            service.submit(task);

        System.out.println(
            "Result = " + future.get());

        service.shutdown();
    }
}
```